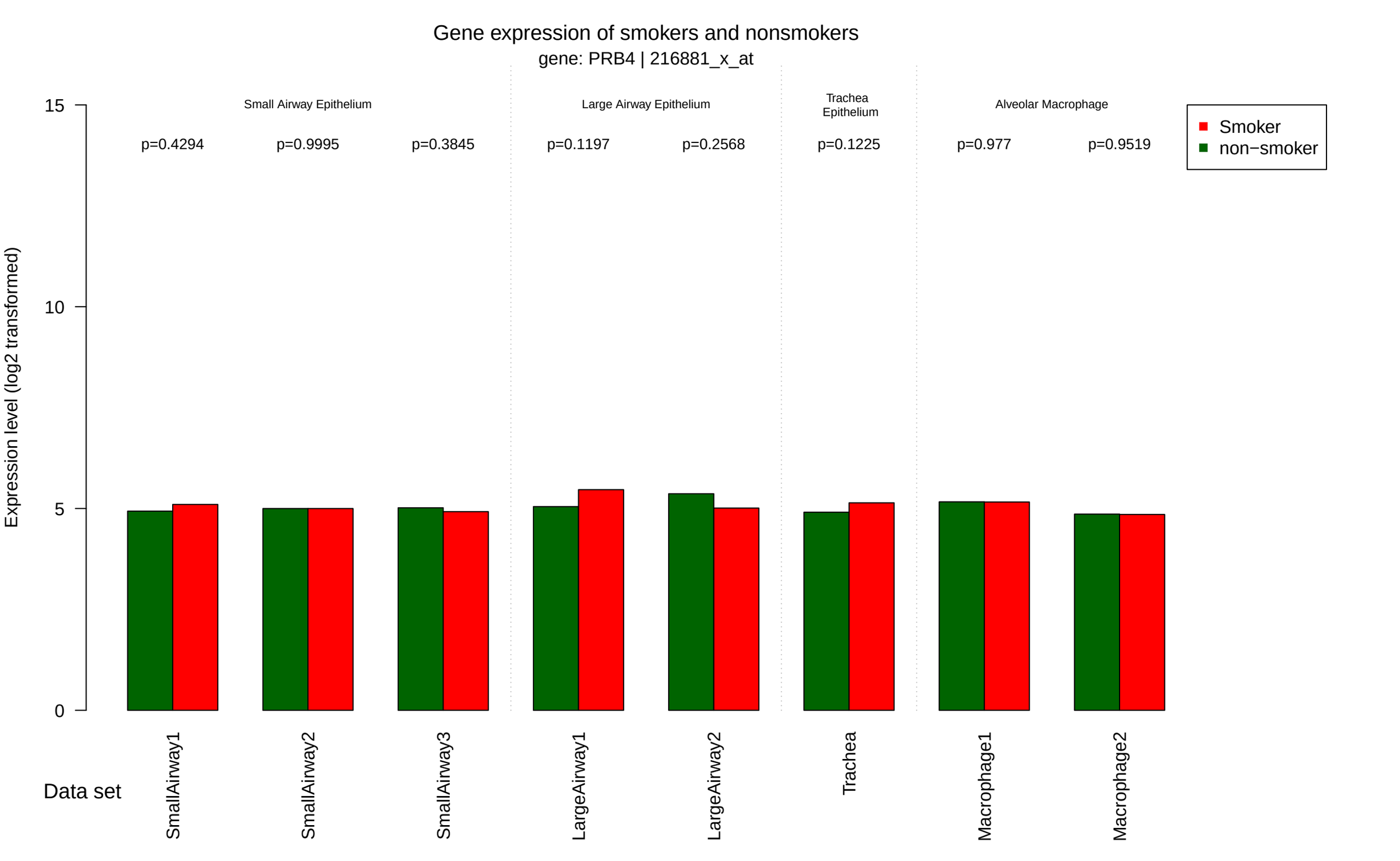
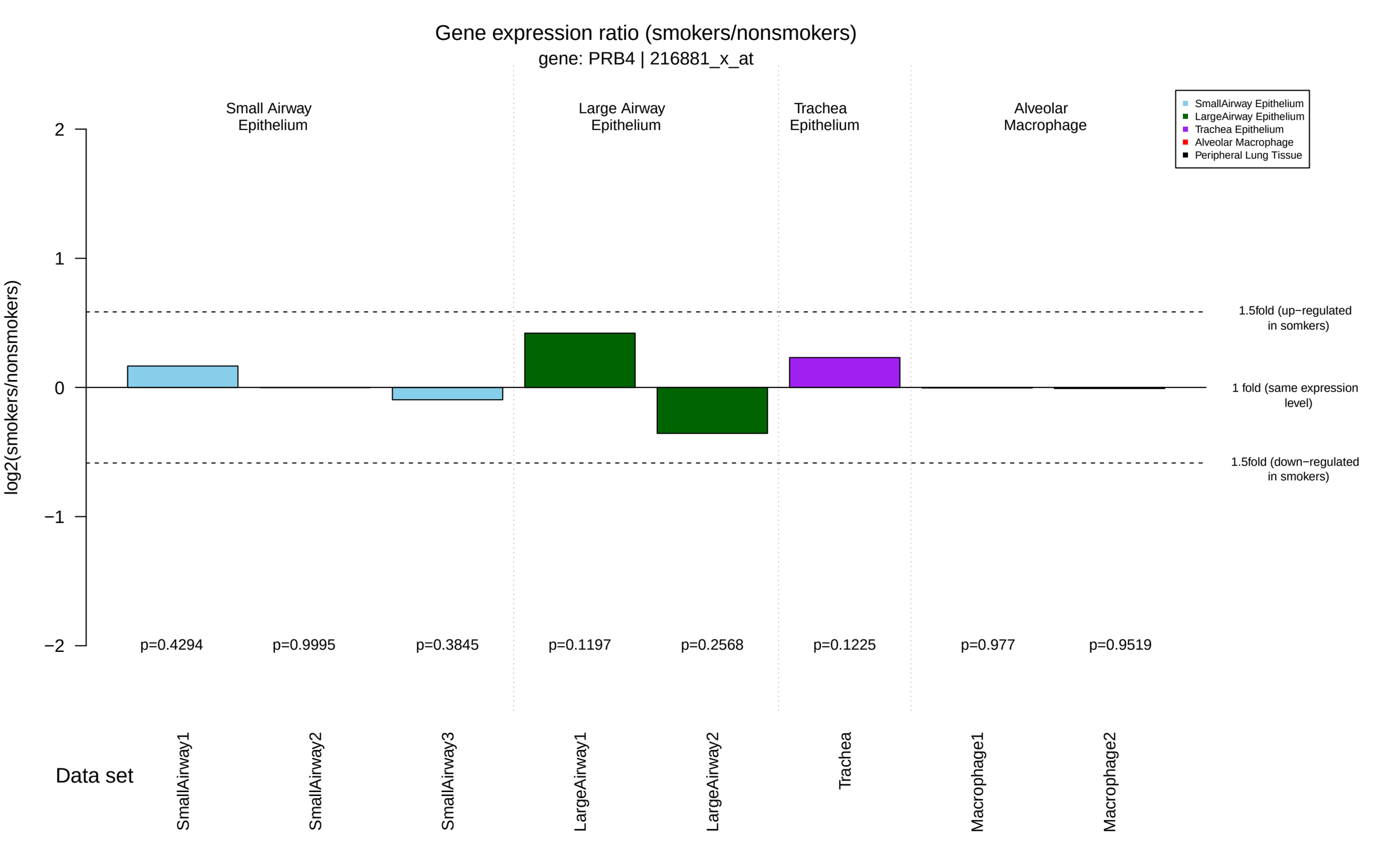
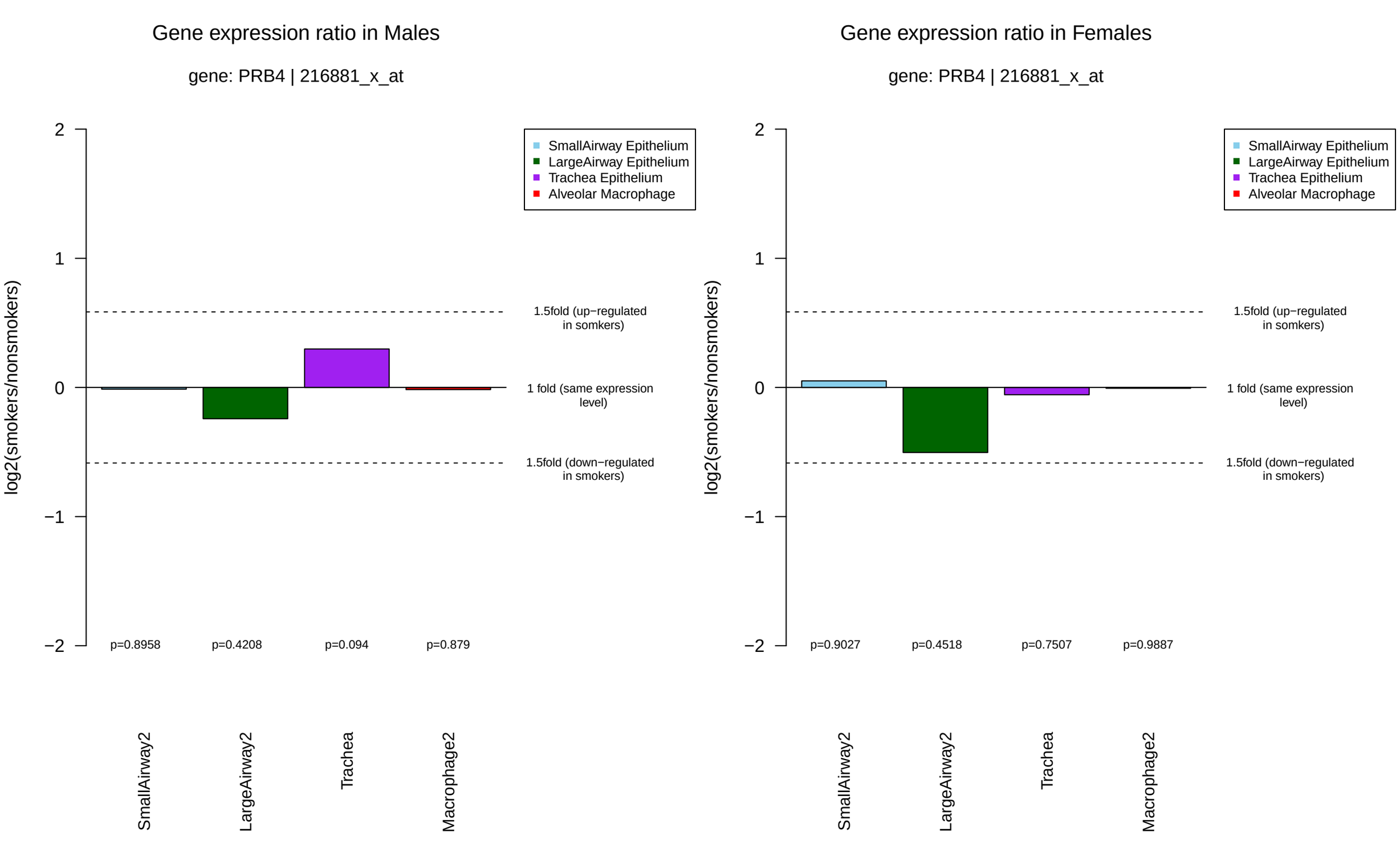
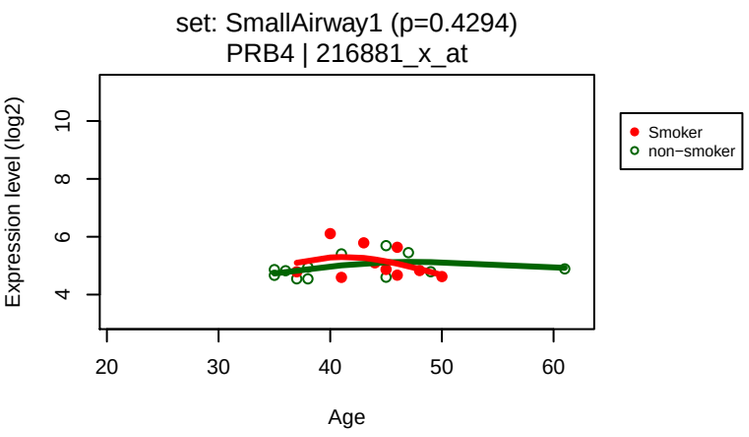
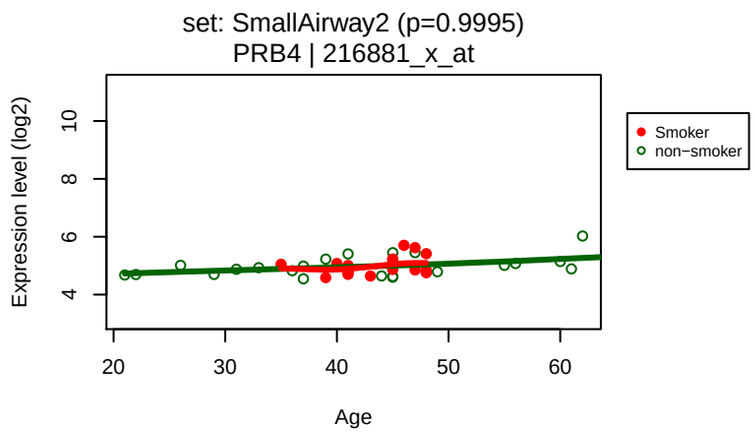
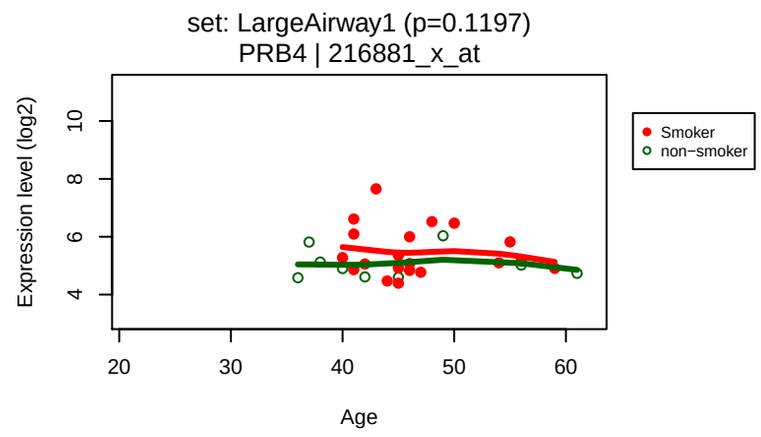

Gene expression of smokers and nonsmokers
gene: PRB4 | 216881_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4294
p=0.9995
p=0.3845
p=0.1197
p=0.2568
p=0.1225
p=0.977
p=0.9519
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

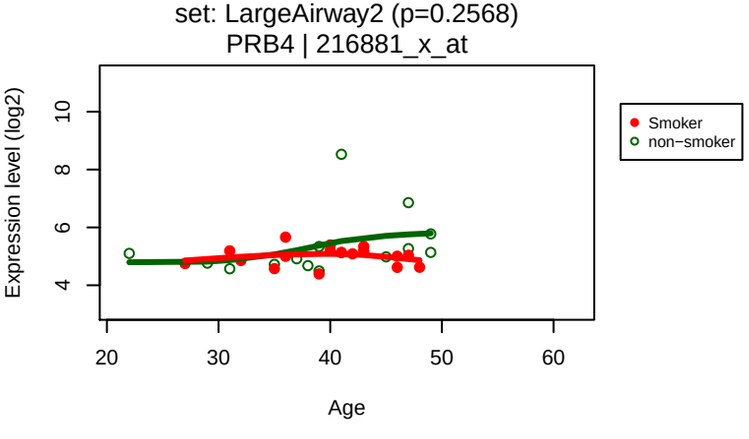

Gene expression ratio (smokers/nonsmokers)
gene: PRB4 | 216881_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4294
p=0.9995
p=0.3845
p=0.1197
p=0.2568
p=0.1225
p=0.977
p=0.9519
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

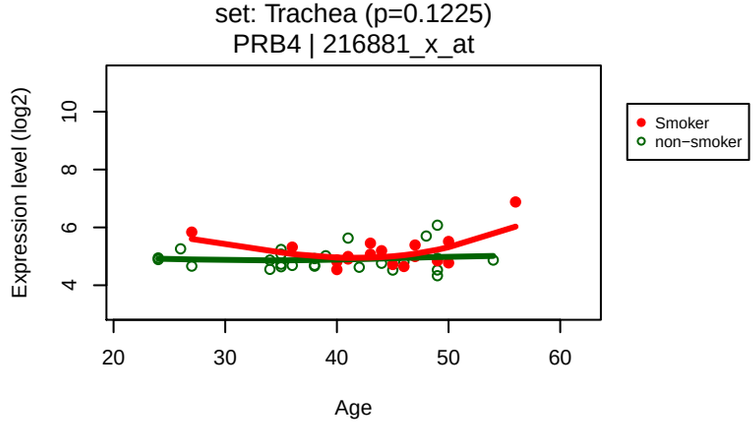

Gene expression ratio in Males
gene: PRB4 | 216881_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8958
p=0.4208
p=0.094
p=0.879
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRB4 | 216881_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9027
p=0.4518
p=0.7507
p=0.9887
SmallAirway2
LargeAirway2
Trachea
Macrophage2

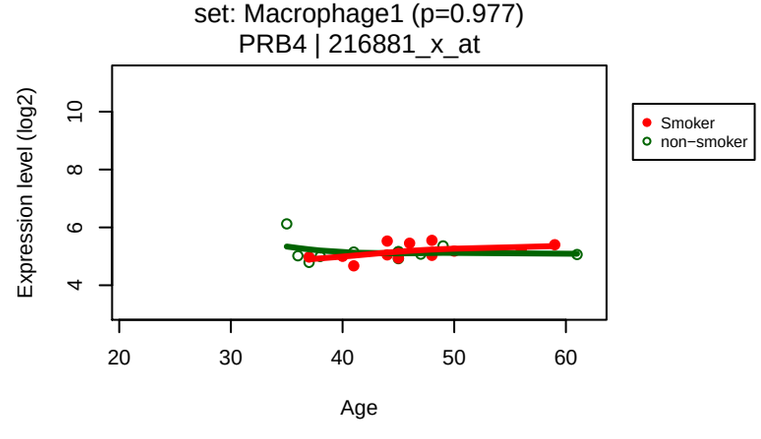

set: SmallAirway1 (p=0.4294)
PRB4 | 216881_x_at
Expression level (log2)
Age
Smoker
non−smoker

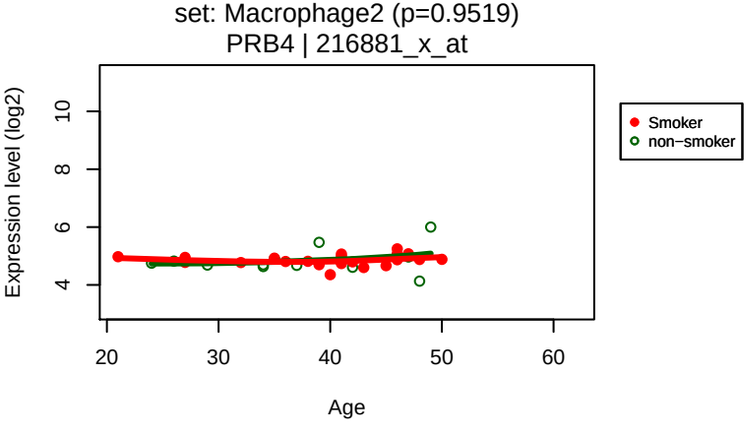

set: SmallAirway2 (p=0.9995)
PRB4 | 216881_x_at
Expression level (log2)
Age
Smoker
non−smoker

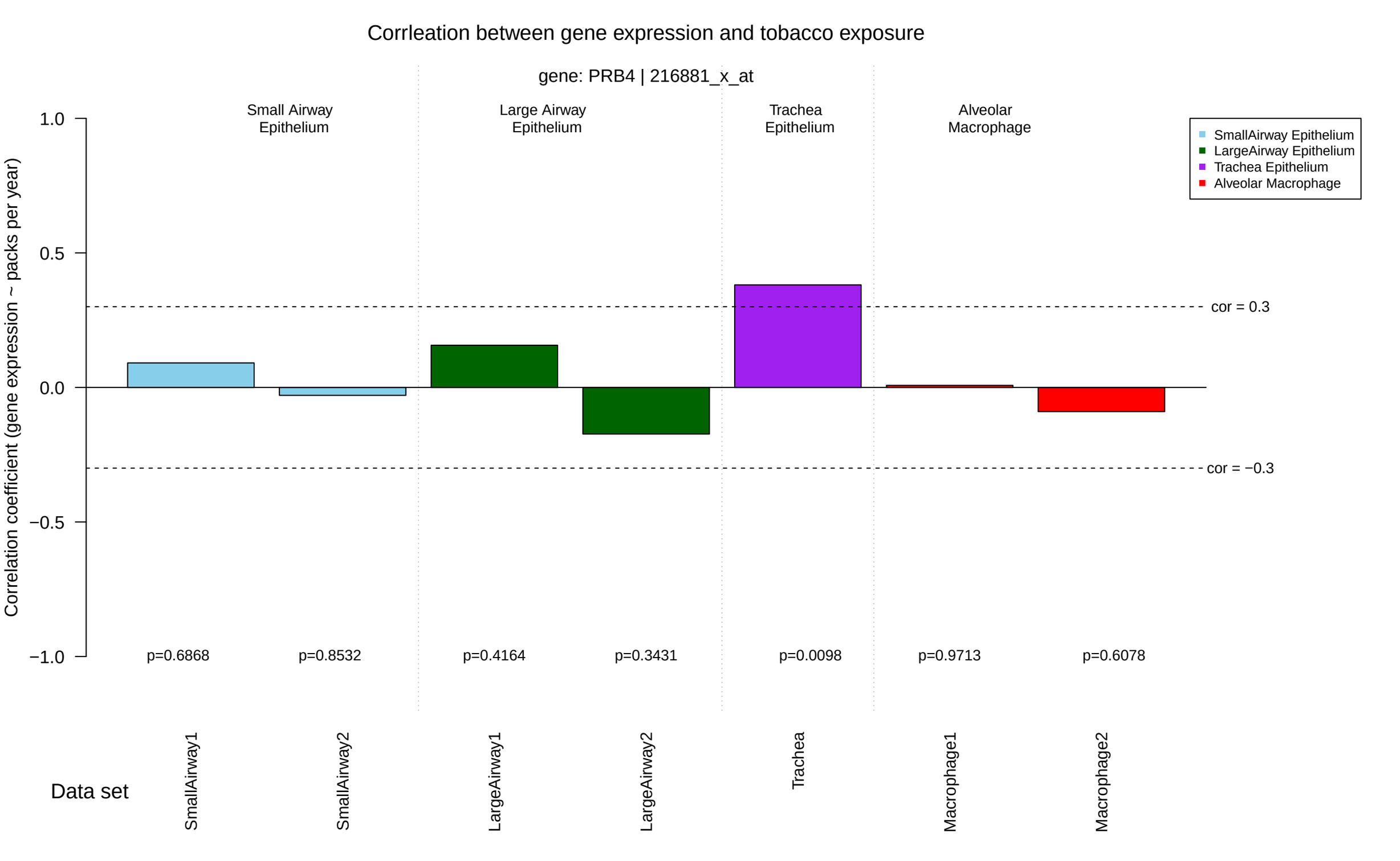

set: LargeAirway1 (p=0.1197)
PRB4 | 216881_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2568)
PRB4 | 216881_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1225)
PRB4 | 216881_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.977)
PRB4 | 216881_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9519)
PRB4 | 216881_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRB4 | 216881_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6868
p=0.8532
p=0.4164
p=0.3431
p=0.0098
p=0.9713
p=0.6078
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2